

Quiz #5

1. The correlation between the heights of husbands and wives in the U.S. is approximately one of the following:

$$-0.9, \quad -0.3, \quad 0.3, \quad 0.9.$$

Which value is most reasonable? Briefly explain your answer.

2. For women age 25 and over in the U.S. in 2005, the relationship between age and educational level (years of schooling completed) can be summarized as follows:

$$\begin{aligned} \text{average age} &\approx 50 \text{ years}, & \text{SD} &\approx 16 \text{ years}, \\ \text{average educational level} &\approx 13.2 \text{ years}, & \text{SD} &\approx 3.0 \text{ years}, \\ r &\approx -0.20. \end{aligned}$$

True or false: As you get older, you become less educated.

Explain your answer.

3. An investigator collected data on the heights and weights of college students. The results are summarized below:

Variable	Average	SD
Men's height	70 inches	3 inches
Men's weight	144 pounds	21 pounds
Women's height	64 inches	3 inches
Women's weight	120 pounds	21 pounds

The correlation coefficient between height and weight for the men was about 0.60; for the women, it was about the same.

If the men and women are combined into one group, would the correlation between height and weight be

somewhat lower than 0.60, about 0.60, or somewhat higher than 0.60?

Choose one option and explain briefly.

4. Suppose a pandas DataFrame named `students` contains two columns: `study_hours` and `exam_score`. Some rows may be missing one of these two values.

Complete the code below so that the same complete pairs are used to make a scatterplot and compute the Pearson correlation coefficient.

```
pair = students[["study_hours", "exam_score"]]._____()
```

```
pair._____(x="study_hours", y="exam_score")
```

```
r = pair["study_hours"]._____(pair["exam_score"])
```

- (a) Fill in the three blanks.
- (b) Why is it useful to make the scatterplot before interpreting r ?
- (c) Suppose the code gives $r = 0.68$. Interpret this value in one sentence. Would this value alone establish that studying more causes higher exam scores?